

Advanced Machine Learning

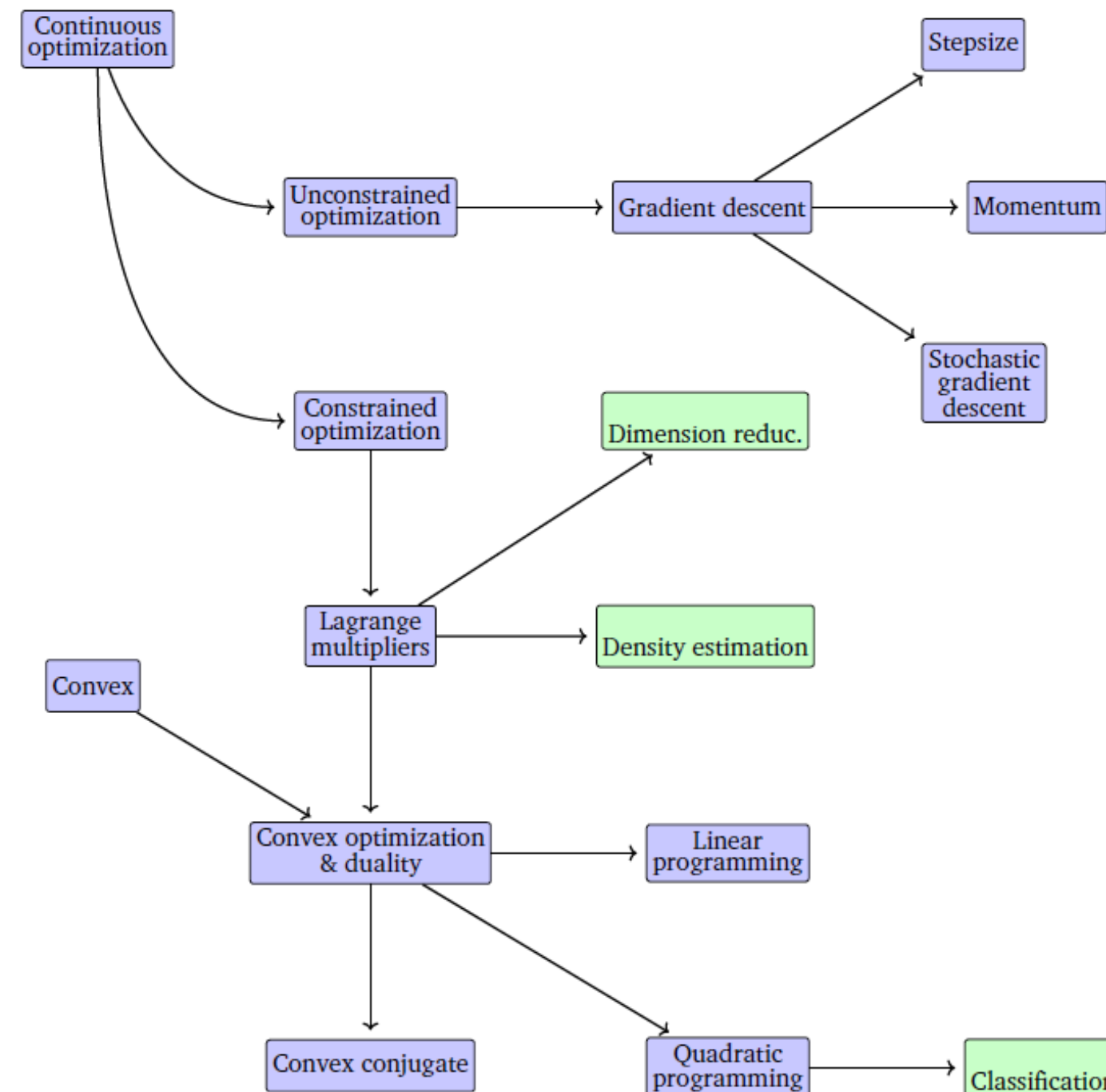
DATA 642

Lecture 2 — Introduction to optimization

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth

Motivation

- Training a machine learning model is essentially a problem of finding a good set of parameters. The notion of "good" is determined by the cost or objective function or the probabilistic model.



Optimization Basics

Cost Function and Objective Function

- Optimization problems are posed by constructing a cost function or objective function $J(\boldsymbol{\theta}) : S \rightarrow \mathbb{R}$ and then finding $\hat{\boldsymbol{\theta}}$ such that

$$J(\hat{\boldsymbol{\theta}}) = \min_{\boldsymbol{\theta} \in S} J(\boldsymbol{\theta}). \quad (1)$$

- Typically S is defined as $\mathbb{R}^l$.
- Suppose you're a manufacturer aiming to maximize profits. Instead of directly maximizing profits, you could minimize the negative of your profit function. For example, if your profit function is $J(\theta)$, where θ represents production levels or pricing strategies, you'd minimize $-J(\theta)$ to find the optimal values.

An example

Let's illustrate how optimization is crucial in linear regression, which is another fundamental machine learning algorithm. In linear regression, we aim to find the best-fitting linear relationship between the input features $\mathbf{x}$ and the target variable y .

Fitting Models to Data:

In linear regression, we model the relationship between the input features ($\mathbf{x}$) and the target variable (y) using a linear equation:

$$y = \boldsymbol{\theta}^T \mathbf{x} + \epsilon \quad (2)$$

where θ represents the parameters (weights) of the model, and ϵ represents the random error term.

During training, we aim to find the optimal parameters θ that minimize the mean squared error (MSE) over the training dataset:

$$J(\boldsymbol{\theta}) = \frac{1}{2m} \sum_{i=1}^m (\boldsymbol{\theta}^T \mathbf{x}^{(i)} - y^{(i)})^2 \quad (3)$$

where m is the number of training examples, $\mathbf{x}^{(i)}$ is the feature vector of the i -th training example, and $y^{(i)}$ is its corresponding target value.

Minimizing Errors

By minimizing the mean squared error (MSE) loss function, we aim to minimize the errors between the predicted values and the actual target values in the training dataset. Optimization helps in fine-tuning the model parameters to reduce these errors and improve the accuracy of predictions.

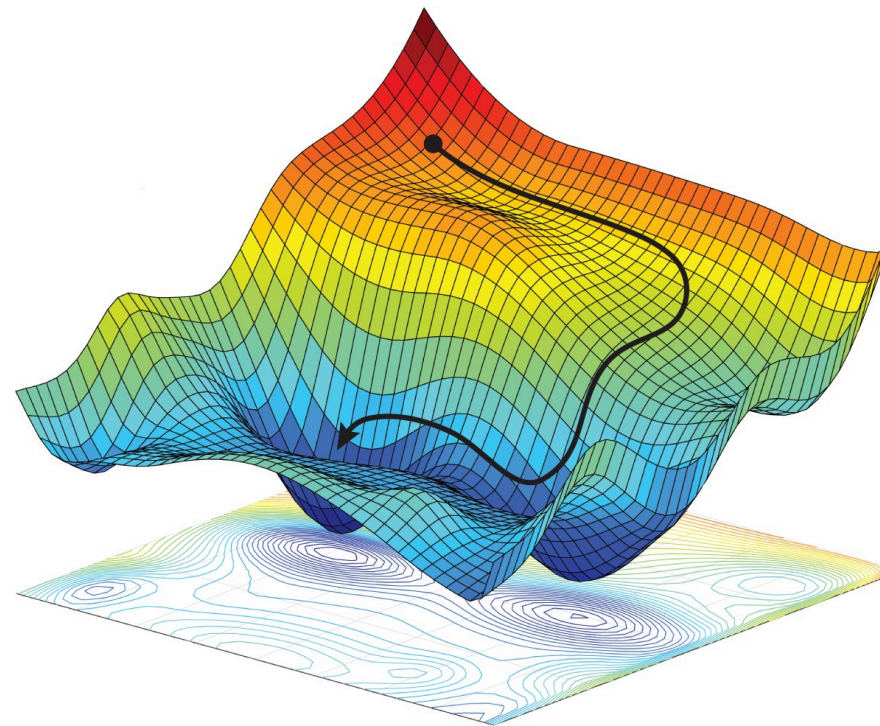
Improving Predictions

As we optimize the parameters θ , the linear regression model learns to make better predictions on both the training data and new unseen data. By iteratively adjusting the parameters using optimization algorithms, we aim to improve the model's predictive capabilities and its ability to generalize to unseen data.

Optimization Basics

Local and Global Minima

- Imagine you're hiking on a mountain trail with many valleys and peaks. A local minimum is like being in a valley where the terrain is lower than the immediate surroundings. If you're at the lowest point in the entire region, you're at a global minimum, which is the lowest point in the entire landscape.



- A point $\hat{\theta}$ is a local minimum if there exists a $\epsilon > 0$ such that for any $\theta \in S$ and $\|\theta - \hat{\theta}\| < \epsilon$ implies $J(\hat{\theta}) \leq J(\theta)$. A point $\hat{\theta}$ is a global minimum if $J(\hat{\theta}) \leq J(\theta)$ for any $\theta \in S$.

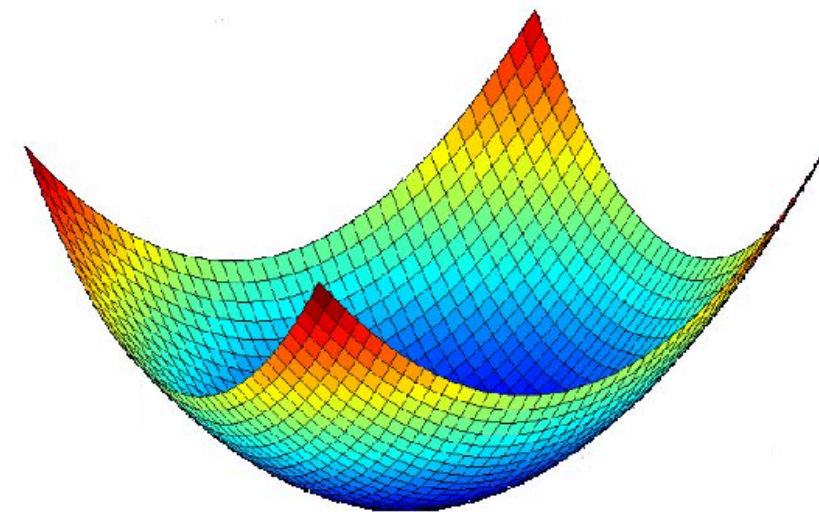
Optimization Basics

Convex Functions

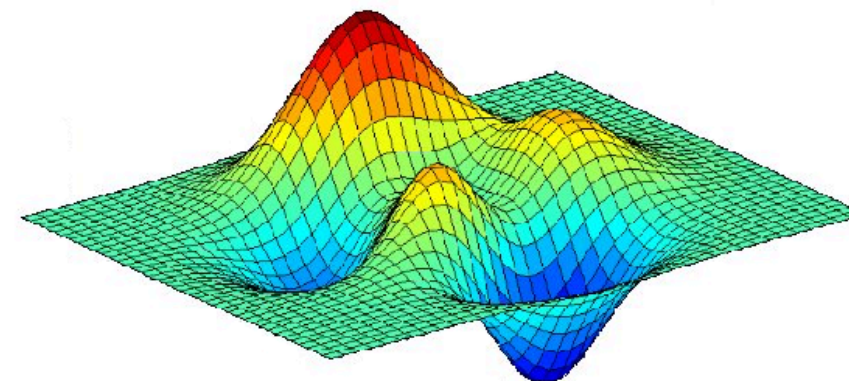
- If J is convex, every local minimum is a global minimum. A convex function is one that satisfies

$$J(t\boldsymbol{\theta} + (1 - t)\boldsymbol{\beta}) \leq tf(\boldsymbol{\theta}) + (1 - t)f(\boldsymbol{\beta}), \quad (4)$$

for all $\boldsymbol{\theta}, \boldsymbol{\beta} \in S$ and all $t \in (0, 1)$.



convex function



non-convex function

Models with Convex Cost Functions

1. Linear Regression:

- The cost function in linear regression, such as the mean squared error (MSE), is convex. This means that finding the global minimum is straightforward, and gradient descent is guaranteed to converge to the global minimum.

2. Logistic Regression:

- In logistic regression, the cost function (log loss or cross-entropy loss) is convex. Similar to linear regression, optimization algorithms can efficiently find the global minimum.

3. Support Vector Machines (SVM):

- SVM aims to find the hyperplane that maximizes the margin between classes. The optimization problem in SVM is convex, allowing for efficient training.

Models with Non-Convex Cost Functions

1. **Neural Networks:**

- Deep neural networks often have non-convex cost functions due to their complex architectures with multiple layers and non-linear activations. Optimization in neural networks can be challenging, and finding the global minimum is not guaranteed.

2. **K-Means Clustering:**

- The cost function in K-means clustering (sum of squared distances to cluster centroids) is non-convex. It involves optimizing the assignment of data points to clusters and the positions of cluster centroids simultaneously, which can lead to local optima.

3. **Gaussian Mixture Models (GMM):**

- GMM is a probabilistic model that assumes data points are generated from a mixture of several Gaussian distributions. Estimating the parameters of GMM involves maximizing the likelihood function, which is non-convex due to the presence of multiple modes.

First-order Condition (Gradient)

- A necessary condition for a point $\hat{\theta}$ to be a local minimum is that the gradient vector $\nabla J(\theta)$ is zero at $\hat{\theta}$. This is called the **first-order condition**.
- Think of climbing a hill. At the peak, you reach a point where there's no uphill direction; you're at a flat spot. In optimization, this corresponds to the gradient being zero. So, at a local minimum, the "slope" of your function is flat.

Second-order Condition (Hessian)

- If a point $\hat{\theta}$ satisfies the first order condition, a sufficient condition for $\hat{\theta}$ to be a local minimum is that the Hessian matrix $\nabla^2 J$ is positive definite. This is called the **second-order condition**.